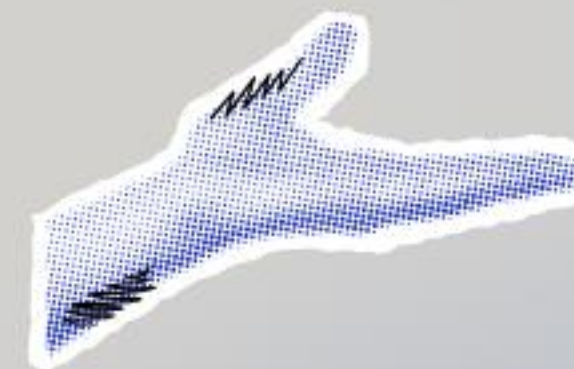


Tell me a joke



Why don't scientists trust atoms?

- Because they make up everything.



>90% = 25 jokes

- Of 1,008 ChatGPT jokes, over 90% were the same 25 memorised ones (Jentzsch & Kersting, 2023)

JOKER Task 4

HUMOUR GENERATION IN EN, FR, ES

- Igor Kuzmin: Pompeu Fabra University
- Task 4 co-organiser
- with Anne-Gwenn Bosser, Jaap Kamps, Liana Ermakova

From memory to mechanism



2025: models detect and translate puns

Strong systems excel at literal translation, but wordplay needs creative re-creation.

The problem: recall, not creation

Open prompts return memorised jokes; word or topic constraints can be met by any joke.

2026: constrain the mechanism

Give the model a pun word and two senses that must both be live in the joke.

The task



The brief

- Pun word + two senses (+ optional keywords)
- lama: Tibetan priest vs. South American animal
- Output: one short humorous text



A valid generation

- Uses the pun word or a clear variant
- Both senses live: remove one, the joke breaks
- Fluent target language, short, self-contained



Why not MWAHAHA?

- SemEval-2026 MWAHAHA constrains words or topic
- Task 4 prescribes the ambiguity mechanism itself
- Judges check the both-senses condition explicitly

Data: 5,335 train and 457 test briefs



English

3,344 train / 150 test

PunEval / ExPUNations with WordNet senses, plus JOKER portmanteau (mots-valises) briefs

Spanish

392 train / 151 test

JOKER 2023 bilingual campaign: only renderings validated as best in their group

French

1,599 train / 156 test

JOKER French corpus: Tom Swifty homophones and general puns with sense annotations

A real brief: barbec-ours



Input at test time

pun_word: barbec-ours

sense_a: barbecue

sense_b: bear (ours)

keywords: BBQ, grill, grizzly, cub

Train-only hint

barbecue + ours

Annotator cues: côtelette, grizzly, cuit. Stripped from the test set together with the reference joke.

Reference joke

fr_0001

"Comment on cuit des côtelettes de grizzly ? Au barbec-ours."

EN: "What do you call grizzly's ribs? A bearbecue."

Evaluation: the Arena



● Constraint-aware judge prompt

Pun word, both senses, naturalness, humour. Not longer, not self-explaining, not position.

● 3 judges, 3 providers, majority

gpt-oss-20b, deepseek-chat, gemini-2.5-flash-lite. Verdict: A, B or tie.

● Sparse Bradley-Terry

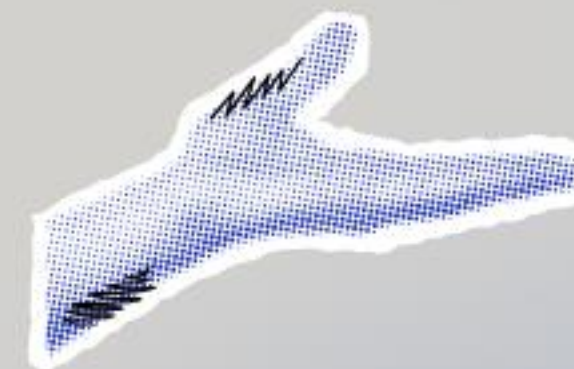
About 40k battles over 365 runs instead of 3.4M pairs.
Bootstrap 95% CIs per track.

4 teams, 365 runs



Winner: SaLT Kimi-CSF

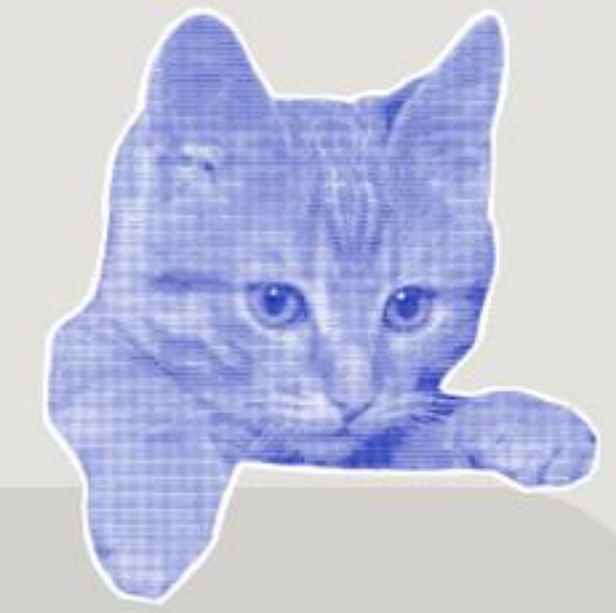
- Kimi = Kimi-K2.6 backbone model
- CSF = Cognitive Synergy Framework from HumorGen (Ajayi & Mitra, 2026)
- K comedic personas each draft a joke; a pairwise ranker keeps the one with the pun word, both senses, natural phrasing, target language
- Ablation: Kimi-Plain (same model, zero-shot) and HumorGen-JOKER 14B/32B (Qwen3 students distilled on CSF data)



Other teams

- UBO: Gemini-3-flash, 4 open LLMs, contrepèteries
- UAmS: open LLMs + humour and pun rerankers
- DUTH: 315 runs, Gemma-2-9B + rerankers

Results: best run per team



English

98 runs, 4 teams

#1 SaLT, 4 runs

BT 1,257, 95% CI [1215, 1308]

SaLT_KimiCSF

#5 DUTH, 80 runs

BT 1,127, 95% CI [1055, 1184]

DUTH_EN_BEST984_PLUS_TRAINSEL_TOP1_B12_04

#6 UBO, 5 runs

BT 1,125, 95% CI [1084, 1168]

UBO_gemini-3_pun

#74 UAms, 9 runs

BT 983, 95% CI [937, 1024]

UAms_Qwen2.5-32B-Instruct

Spanish

134 runs, 4 teams

#1 SaLT, 4 runs

BT 1,515, 95% CI [1450, 1599]

SaLT_KimiCSF

#4 UBO, 5 runs

BT 1,351, 95% CI [1310, 1396]

UBO_gemini-3_pun

#6 DUTH, 120 runs

BT 1,314, 95% CI [1259, 1367]

DUTH_ARENAJUDGE_ES

#60 UAms, 5 runs

BT 1,061, 95% CI [1026, 1105]

UAms_Llama-3-Instruct-Neurona-8b

French

133 runs, 4 teams

#1 SaLT, 4 runs

BT 1,395, 95% CI [1342, 1449]

SaLT_KimiCSF

#2 DUTH, 115 runs

BT 1,304, 95% CI [1245, 1417]

DUTH_FR_LEAK_NOR13_DROP_R04

#7 UBO, 10 runs

BT 1,279, 95% CI [1234, 1341]

UBO_gemini-3-flash_pun

#41 UAms, 4 runs

BT 1,191, 95% CI [1149, 1229]

UAms_Chocolatine-2-14B-Instruct

= rank of the run among all runs in its track. BT = Bradley-Terry rating from the 3-judge majority vote, 95% bootstrap CI.

Arena battles: EN, ES, FR



English

pun word: lama

Sense 1: a Tibetan or Mongolian priest of Lamaism

Sense 2: a wild or domesticated South American cud-chewing animal related to camels but smaller and lacking a hump

A: DUTH

I followed a lama to the mountains, hoping to learn some spiritual enlightenment. He just kept asking me to spit out my gum.

B: SaLT Kimi-CSF

I joined what I thought was a Buddhist retreat—everyone followed our leader religiously. Then I noticed he spat when excited and ate exclusively hay. Turns out my lama was not what I expected.

Judges: gpt-oss-20b B, deepseek-chat B, gemini-2.5-flash-lite B
Ensemble: B (unanimous)

Spanish

pun word: picazón

Sense 1: picazón (démangeaison)

Sense 2: itching ~ alergia; picazón; fiebre del heno

A: SaLT Kimi-CSF

Le dije al alérgico: “Relájate, es solo fiebre del heno.” Él respondió: “¿Solo? Tengo picazón... ¡y ya no sé si es la alergia o el heno existencial!”

B: DUTH

Nadie esperaba que picazón fuera el remate.

Judges: gpt-oss-20b A, deepseek-chat A, gemini-2.5-flash-lite A
Ensemble: A (unanimous)

French

pun word: potentiel

Sense 1: potentiel (électricité)

Sense 2: potentiel (ressources)

A: SaLT Kimi-CSF

Panne électrique. Je sors mon facture, lumière de bougie: “Monsieur, vous avez du potentiel.” Je tremble... est-ce qu’ils parlent de mes ressources, ou est-ce que le compteur se moque de moi?

B: DUTH

Il a un potentiel énorme, mais il préfère rester branché.

Judges: gpt-oss-20b A, deepseek-chat B, gemini-2.5-flash-lite B
Ensemble: B (2–1 split, brevity preferred)

A and B = order shown to the judges. Each judge votes A, B or tie; the ensemble takes the majority.

What worked, what didn't



What worked

- Test-time guidance: Kimi-CSF beat the same model without it in all 3 tracks (ES 1,515 vs 1,437)
- Strong model + plain prompt: UBO's Gemini-3-flash is top 7 in every track
- Generate many, then select: SaLT personas + ranker; DUTH rerankers reach #2 in FR



What didn't

- Distillation alone: Qwen3 14B/32B students stay below Kimi-Plain
- Small models: the same prompts on 7-9B open models land mid-table; flan-t5 is at the bottom
- Compliance alone: UAms has 96.7% pun-word coverage, yet ranks 74 in EN



Scale is not enough

- 315 DUTH runs did not beat 4 SaLT runs
- Contrepèteries judged with a pun prompt sit at the bottom of FR

Limitations



Ranking is LLM-only

Humans evaluated only small samples. LLM judges are indicative but imprecise: Cohen's kappa 0.43-0.53 between judges.

The judge prompt shapes the ranking

A tight one-liner can beat a richer joke (potentiel). French contrepèteries were judged with the pun prompt and sit at the bottom of FR.

Late data, few teams

4 teams submitted. DUTH's 315 runs dominate the pool, and their Task 4 working note is not available yet.

Next: JOKER 2027



Human votes in the Arena

Native speakers vote in each language, alongside the LLM judges.

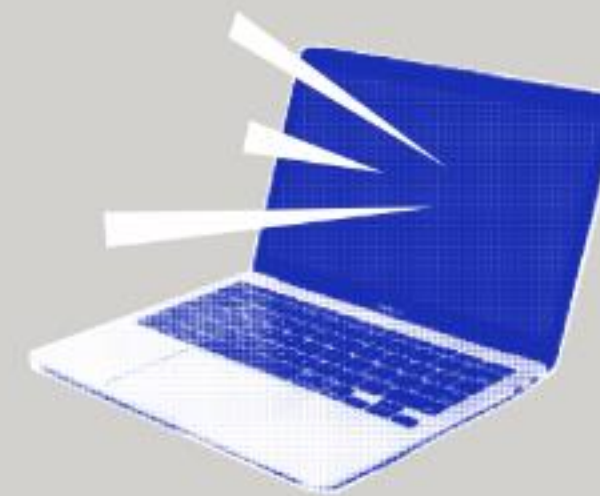
Judges measured against humans

Human-LLM agreement becomes the metric to select and reweight judges. Battles with human and LLM votes released as a dataset.

Novelty check

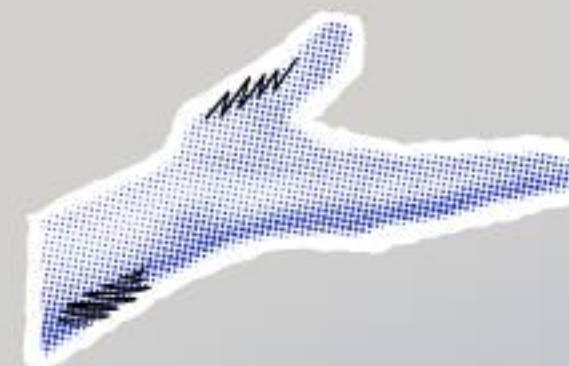
Penalise generations that repeat known jokes: no more 25 recycled jokes.

Thank you! Questions?



Takeaways

- Constraining the mechanism moves models off memorised jokes
- Following the brief is easy; making both senses land is not
- The bottleneck is judging funniness, not generating text
- Next: more judges, prompt sensitivity, form-specific criteria (contrepèteries), shared protocol with MWAHAHA



Join us

- Codabench stays open for post-deadline runs
- Planning Session today, 15:00, Room 131
- CLEF 2027 JOKER Track

Backup: the judge prompt



4 criteria, weighed together

1 pun word or a clear variant; 2 both senses live, removing either breaks the joke; 3 fluent target language; 4 lands as funny: a short punchline beats self-explaining wordplay

Anti-bias rules

Do not prefer the longer, the more formatted or the self-explaining answer; ignore A/B order. If both fail criteria 1 or 2, prefer the funnier surface text.

Output

2-4 sentence comparison, then [[A]], [[B]] or [[C]] (comparable). Prompt pair-pun-v1, adapted from MT-Bench pair-v2 and the PunEval pun-explanation judge.